



Basics of

Brain-Computer Interface

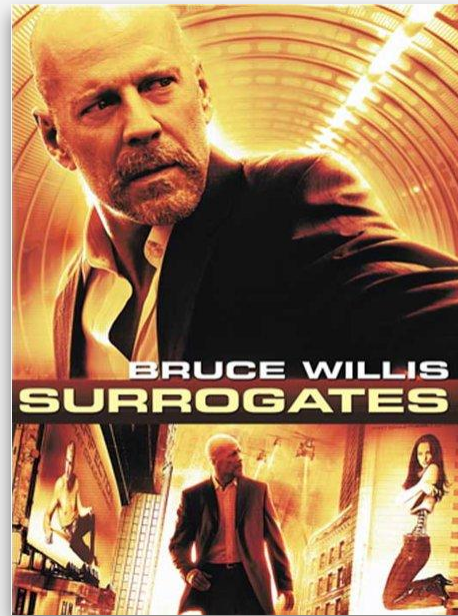
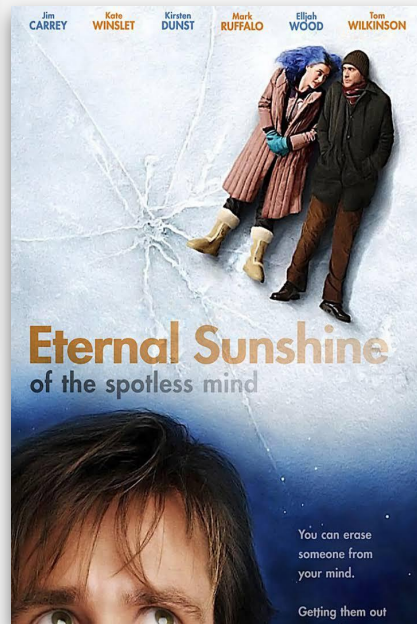
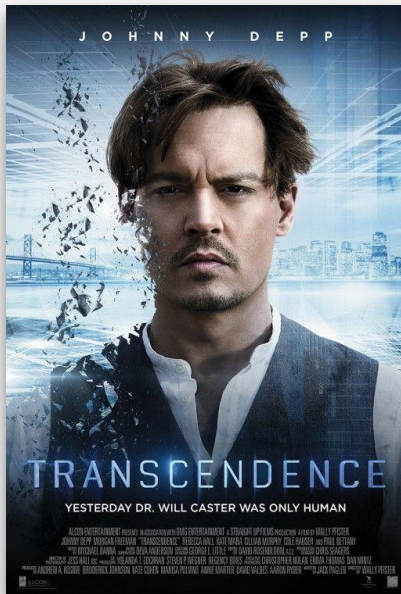
New Technological Innovation Webinar Series

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Contents

- Think beyond the reality
- BCI applications
- Popularity
- What is BCI?
- Technologies
- What is EEG and why?
- Role of machine learning
- Challenges



BCI applications



- Controlling robots (2017)
 - EEG headset is used to collect brain data.
 - After recognizing the recorded pattern of the mental command, sends it to the robot arm.

BCI applications (cont.)



- Controlling drones (2015)
 - This application utilizes a wireless portable EEG headset to send metal commands to the drone.

BCI applications (cont.)

- BCI speller and mouse (2011)
 - Visual cues are used to stimulate the brain and get a clear mental command to type a character or move the mouse cursor.



BCI applications (cont.)



- Image reconstruction using brain data (2017)
 - fMRI images are used to train a machine learning model.
 - Trained model is able to reconstruct an image for the given fMRI data.

BCI applications (cont.)



- Brain-to-vehicle by Nissan (2018)
 - Nissan has developed a BCI application to speed up reaction time for drivers.
 - The system uses portable EEG headset to capture brain signals.

BCI applications (cont.)



- BCI with VR (2016)
 - In some applications, mental commands are used to control things in the virtual world
 - Also, brain data can be used to recognize emotional state.

BCI applications (cont.)



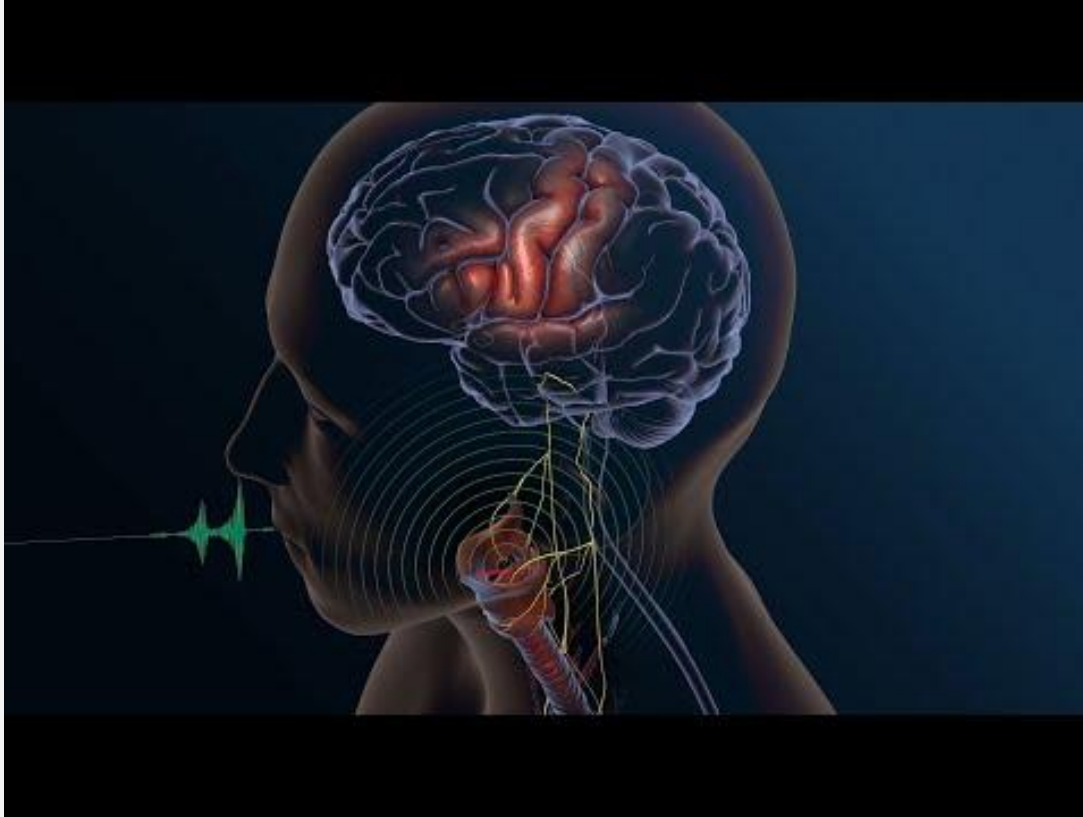
- BCI exoskeleton (2015)
 - Usually exoskeletons are for reinforce human body movements.
 - In this application, the exoskeleton is been controlled by mental commands using EEG headset.

BCI applications (cont.)



- BCI in field of education
 - Portable EEG headset is used to track focus level of students.

BCI applications (cont.)



- Speech from brain signals (2019)
 - A system that can directly translate brain signals into human-understandable speech.

BCI applications (cont.)



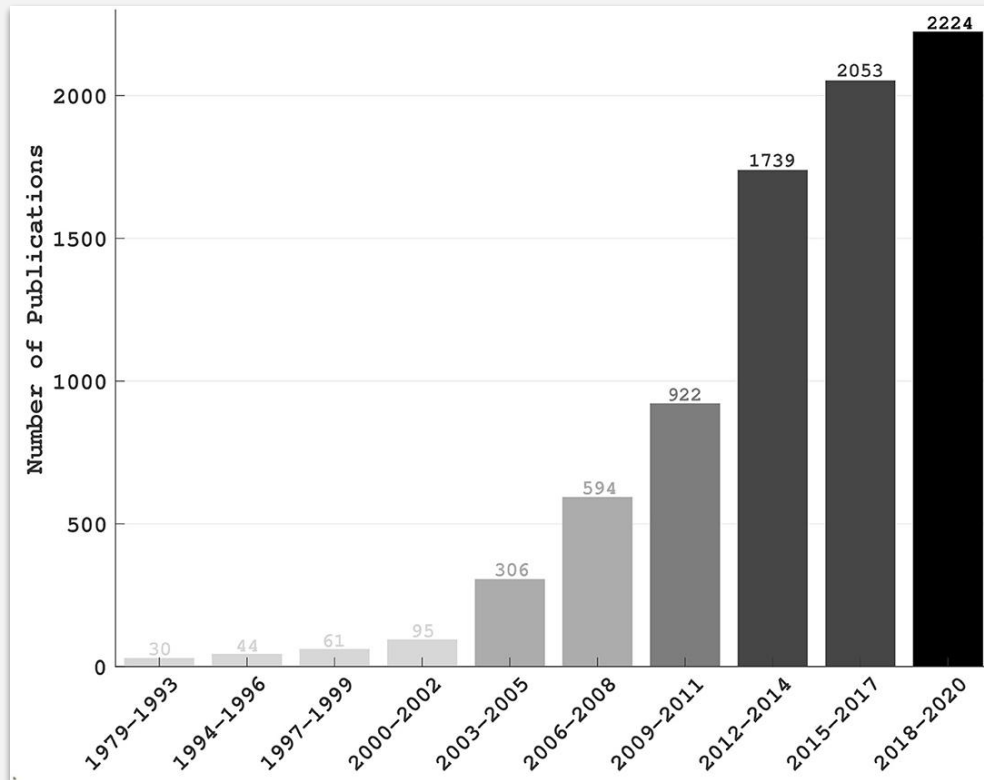
- Brain-to-text system (2021)
 - This system directly translates brain signals into human readable text.



BCI applications (cont.)

- Brain signal-based authentication system
- Epilepsy seizure detection
- Meditation applications
- Emotion classification
- Wheelchair control
- Sleep disorder studies
- etc.

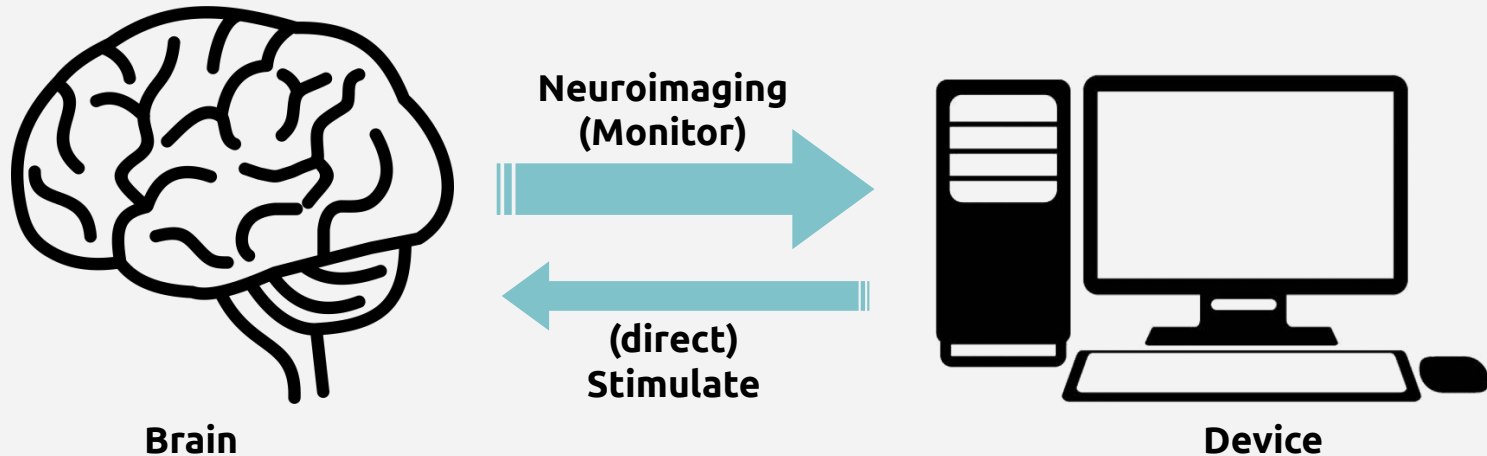
Popularity



Source: <https://doi.org/10.3389/fnsys.2021.578875>

What is BCI?

- Direct communication between enhanced brain and an external device.
- Subcategory of Human-computer interaction (HCI)





Related subject areas

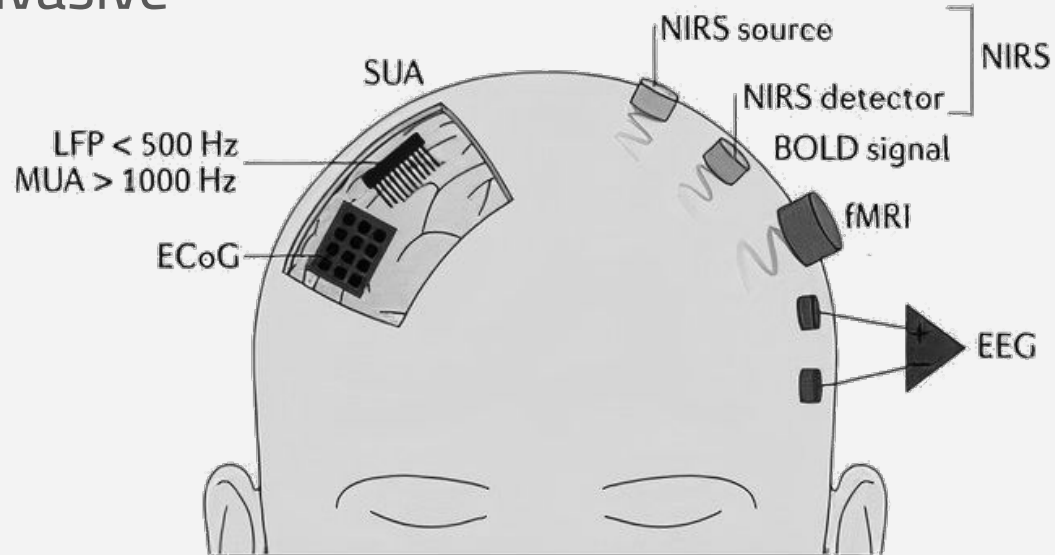
- **Brain anatomy:** study of brain structure
- **Neuroscience:** study of structure or function of the nervous system and brain

What computer scientist/engineers familiar with

- **Digital signal processing (DSP):** process signals digitally using computing devices
- **Digital image processing (DIP):** process images digitally using computing devices
- **Machine learning (ML):** ability to find patterns in data

Interacting with brain

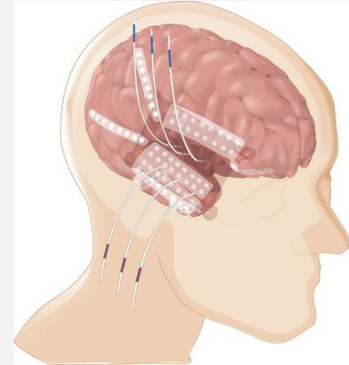
- Methods
 - Non-invasive
 - Partially invasive
 - Invasive



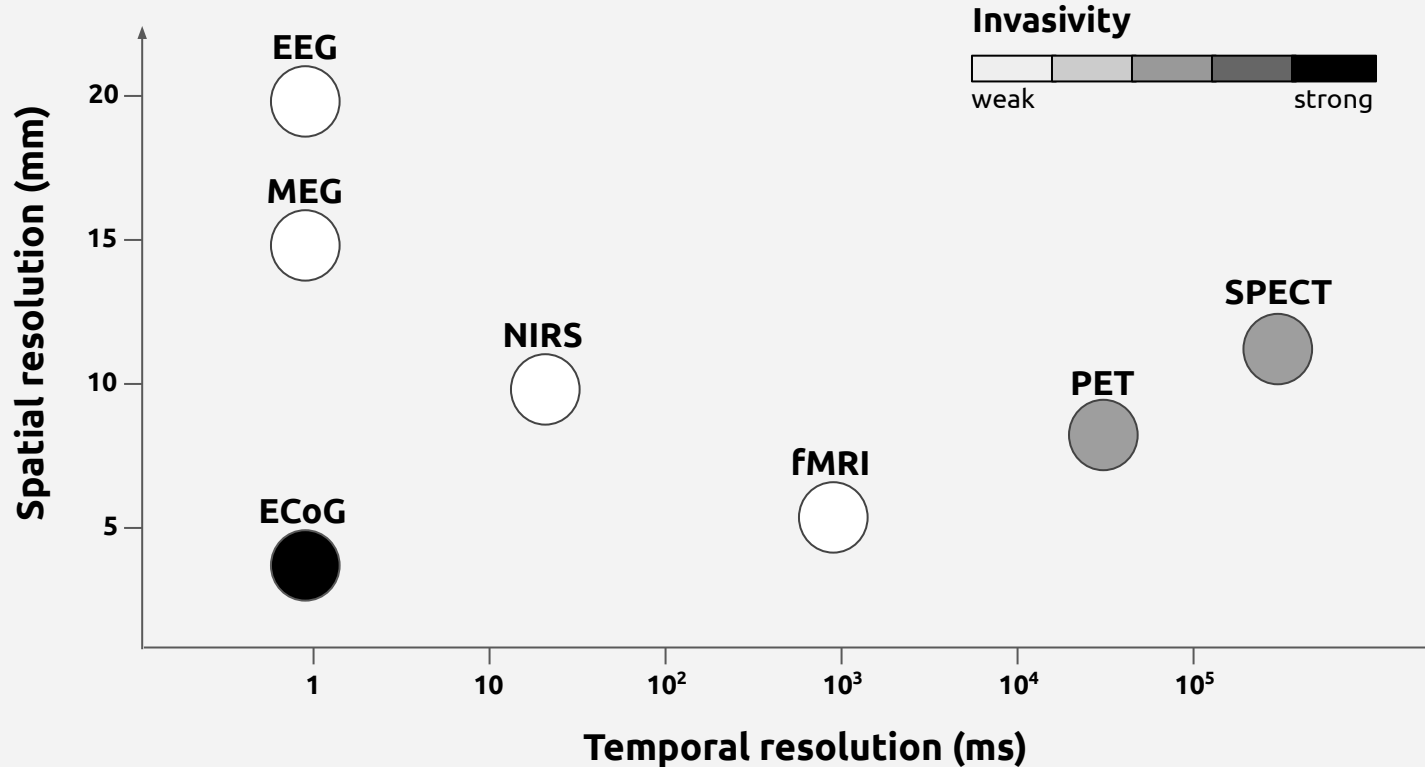
Source: <https://www.nature.com/articles/nrneurol.2016.113>

Brain activity monitoring techniques

- **fMRI:** Functional Magnetic Resonance Imaging
- **EEG:** Electroencephalography
- **MEG:** Magnetoencephalography
- **NIRS:** Near-infrared spectroscopy
- **ECoG:** Electrocorticography



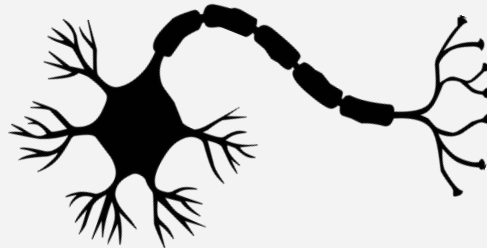
Spatial and temporal resolution of neuroimaging



Source: <https://tel.archives-ouvertes.fr/tel-01175851/>

What is EEG and why EEG?

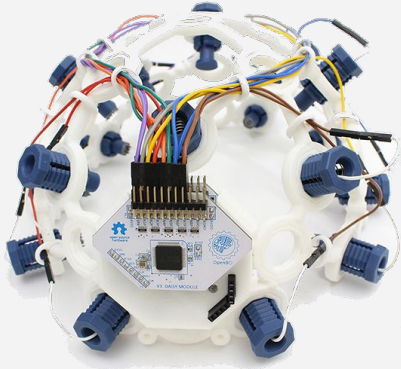
- Electroencephalography (EEG): Electrophysiological monitoring method to record electrical activity of one's brain
- Subtle electric field is generated by neurons
- EEG electrodes can capture the electric field of group of neurons
- Portable, non-invasive, affordable, and easy to use



EEG devices

- Clinical- and consumer-grade devices are available
- Two types of electrodes, Wet and dry
- Has different number of electrodes

OpenBCI



Emotiv EPOC+



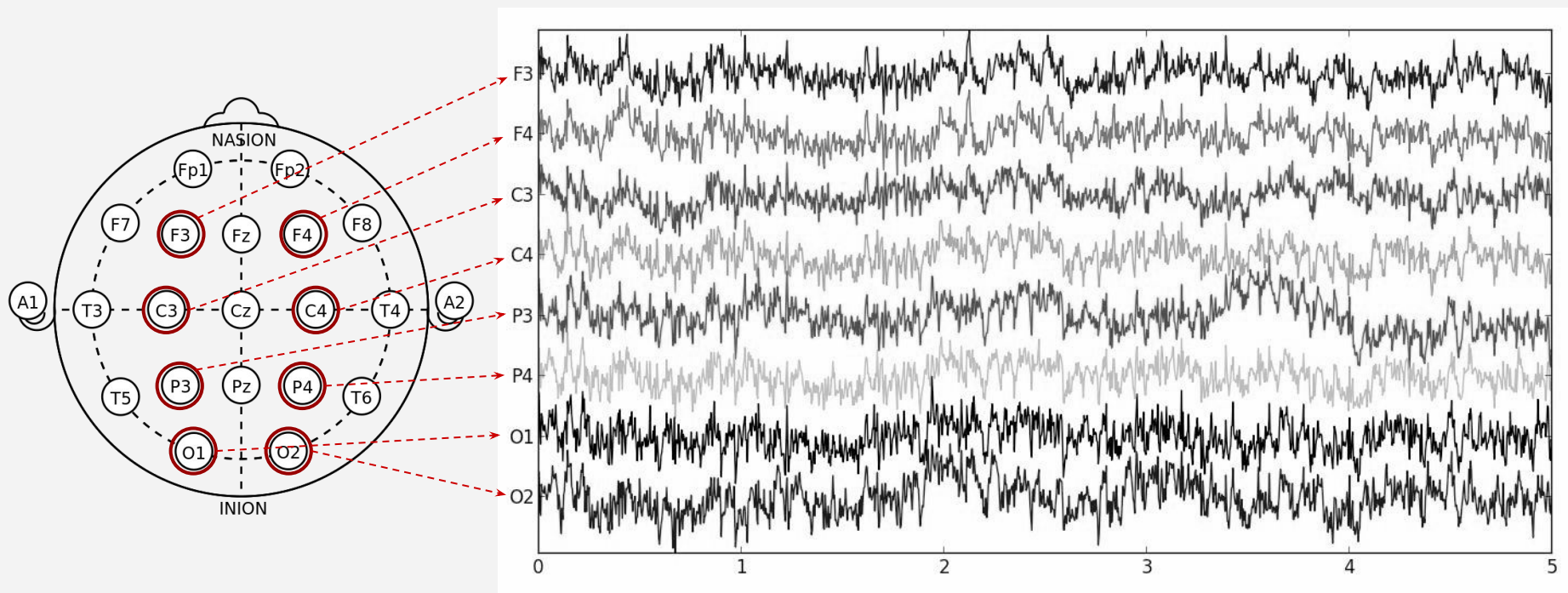
Emotiv Insight



Muse

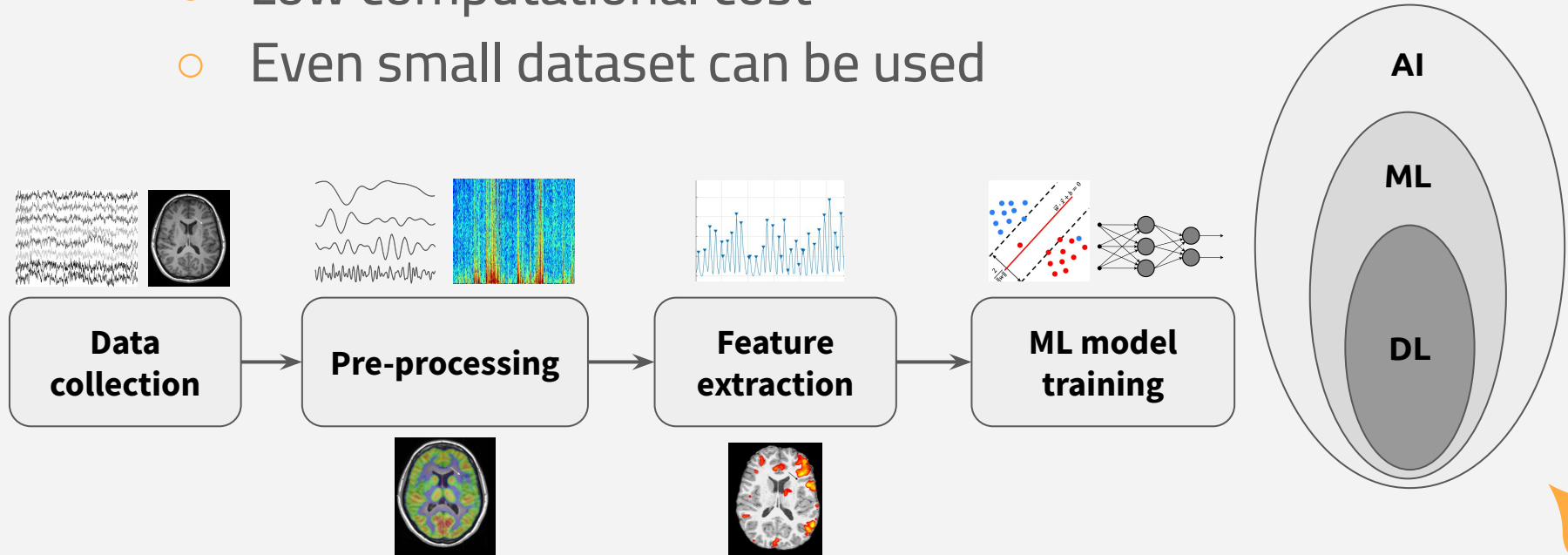


Sample EEG signal



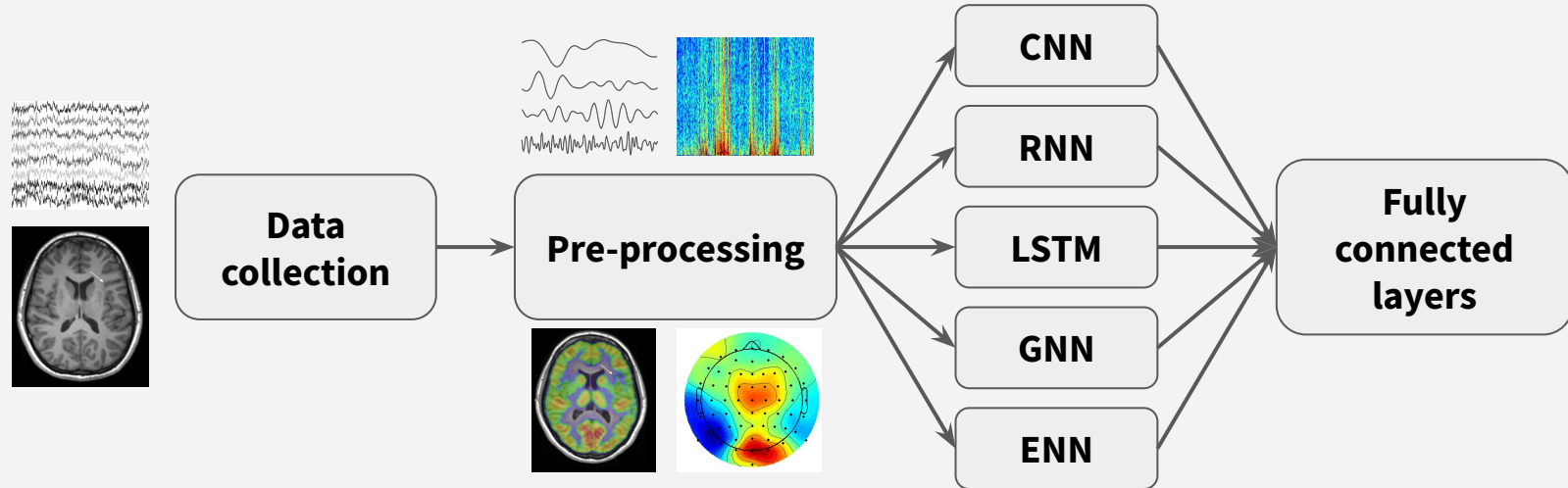
Role of conventional machine learning in BCI

- Traditional ML techniques (LDA, SVM, kNN, ANN)
 - Feature extraction is needed
 - Low computational cost
 - Even small dataset can be used



Role of deep learning in BCI

- No need to extract features
- Need a large dataset
- High computational cost
- Transfer learning can be used in some cases





Challenges

- Uncertainty of brain patterns
- Difficulty of data collection
- Unreliability of data quality
- Misbeliefs about BCI technologies

Can we control brain with a device?

- Repetitive Transcranial Magnetic Resonance (rTMS)
- Transcranial Direct Current Stimulation (tDCS)

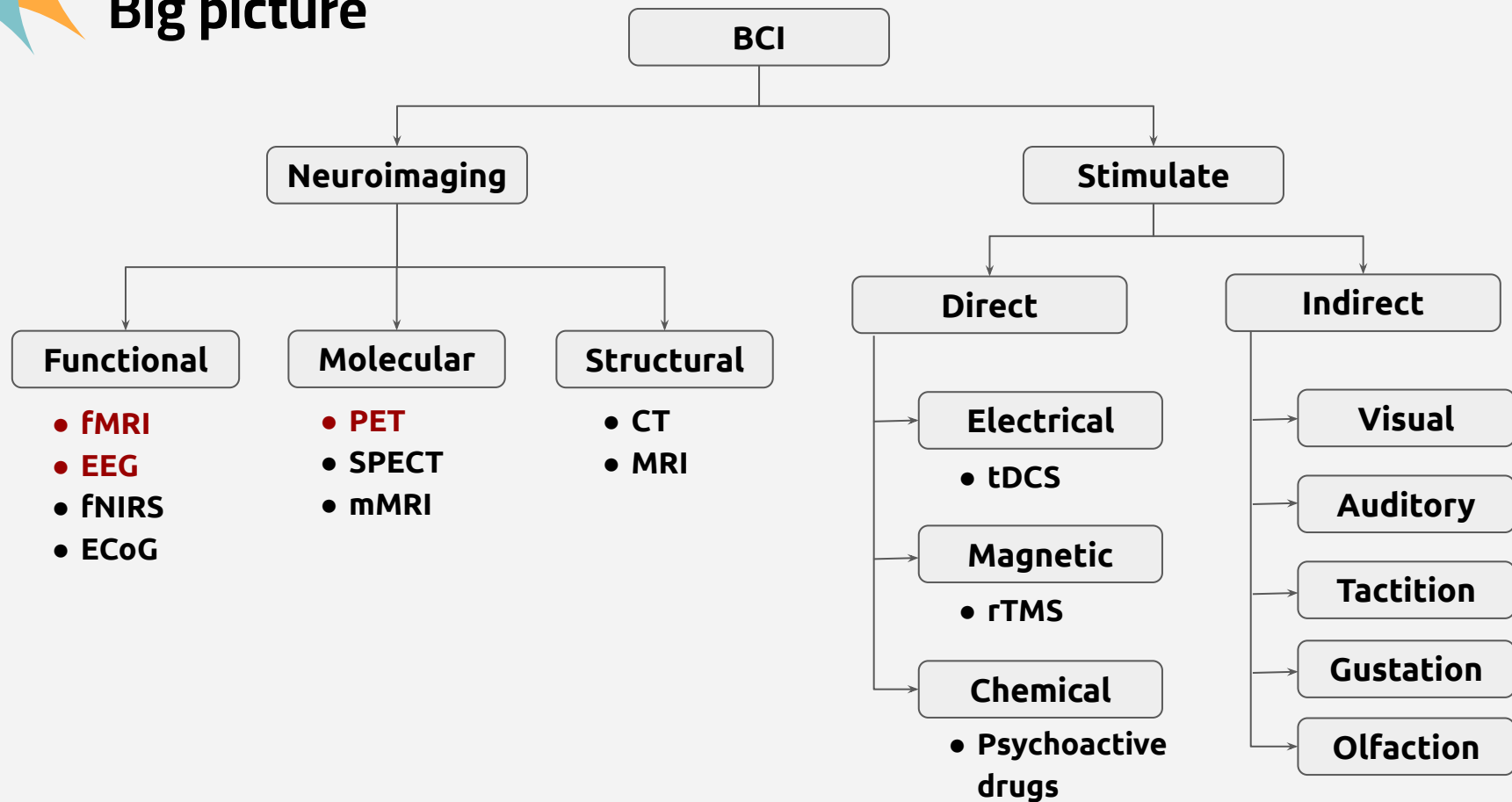


Applications on brain stimulation

- Effect of rTMS on speech (2011)



Big picture

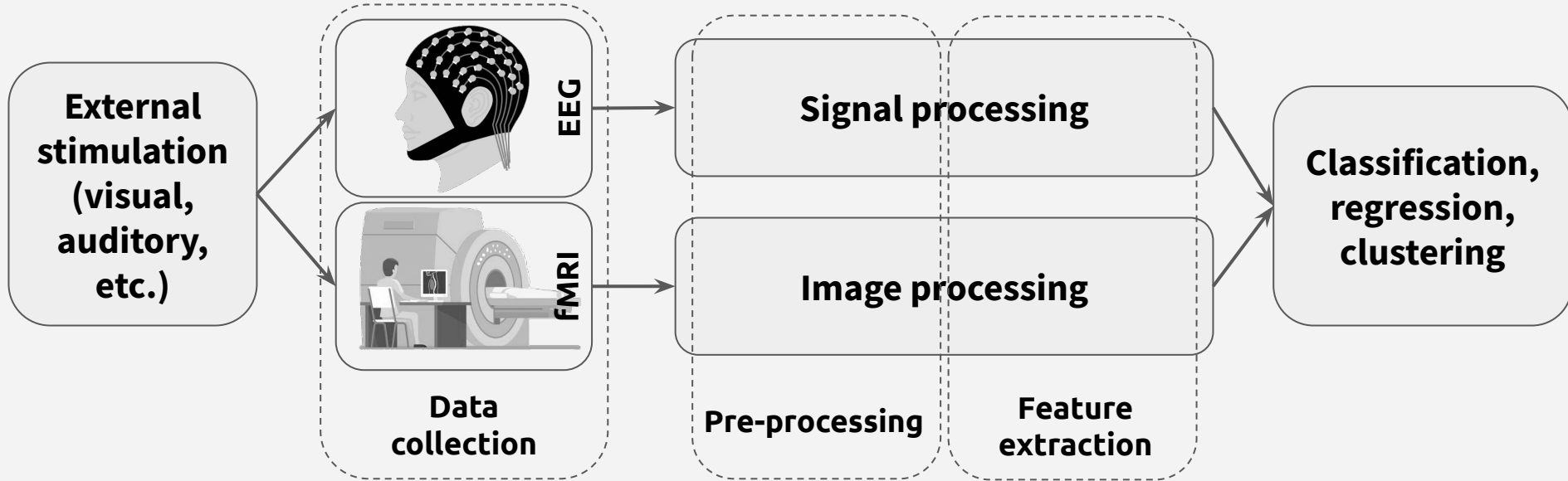


Thank you!

Any questions?

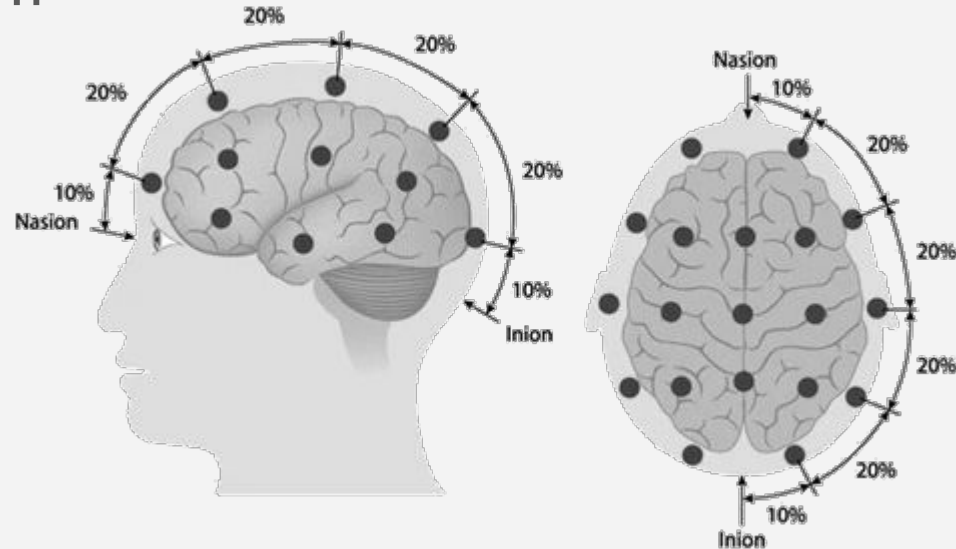
Blog: www.i-linkz.net

Basic flow of most BCI systems



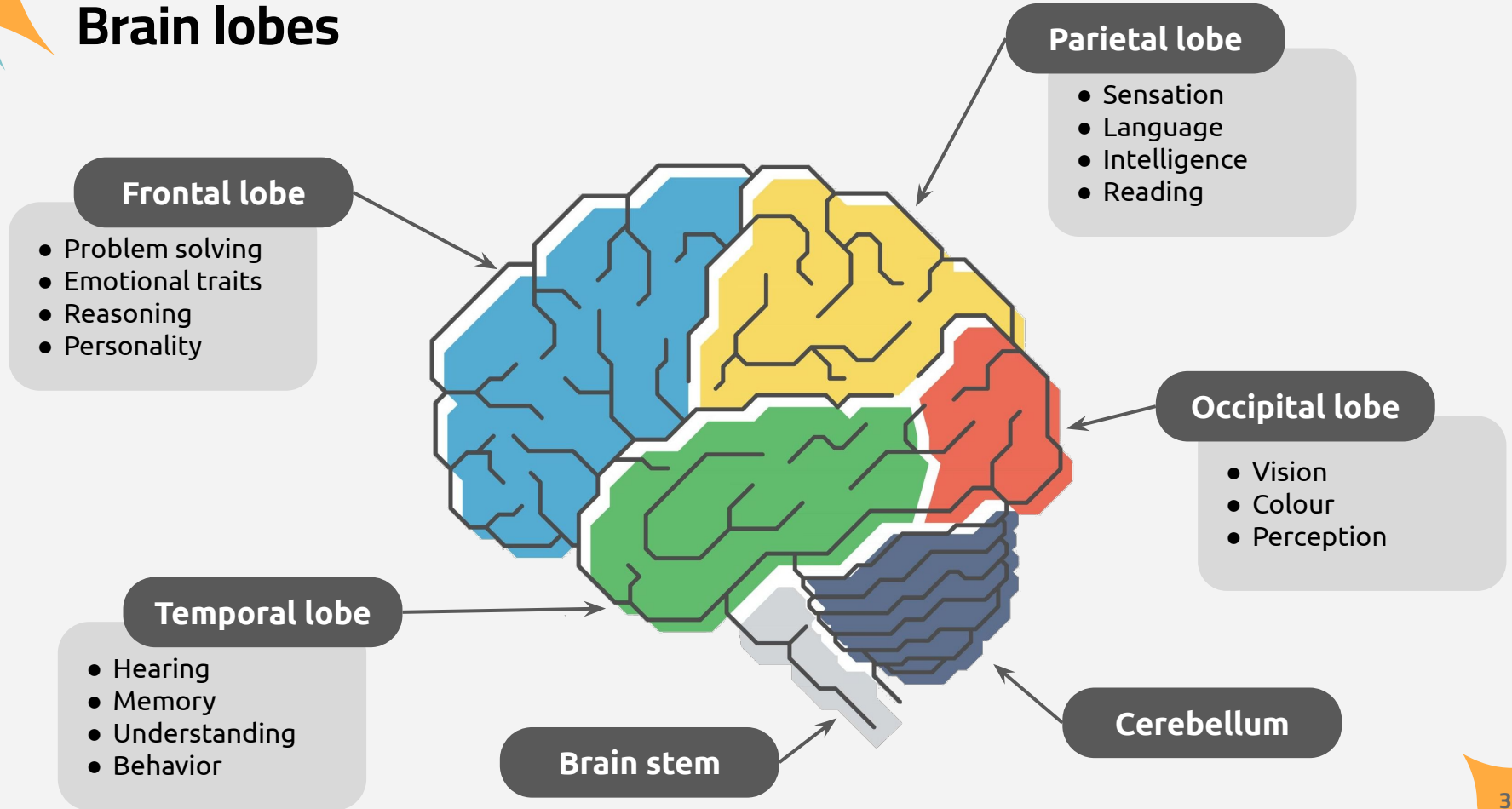
Electrode placements of EEG systems

- 10-20 system
- 10-10 system or 10%
- 10-5 system



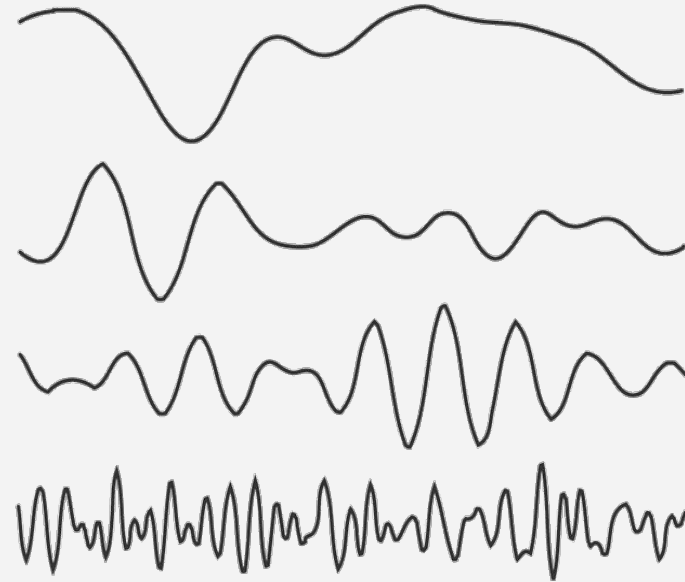
Source: <https://doi.org/10.1186/s13634-015-0251-9>

Brain lobes



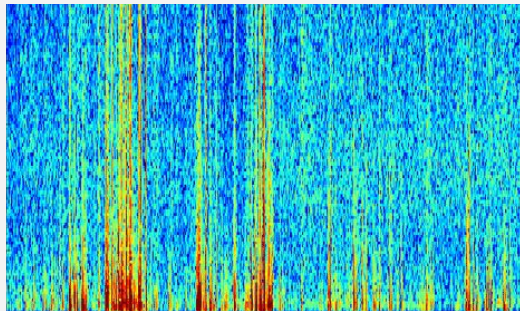
EEG frequency bands

- Delta: 1 - 4 Hz
 - Highest amplitude
 - Presents during deep sleep
- Theta: 4 - 8 Hz
 - Mental and physical relaxation
- Alpha: 8 - 13 Hz
 - Focused attention, memory recall
- Beta: 13 - 30 Hz
 - Active thinking, active concentration



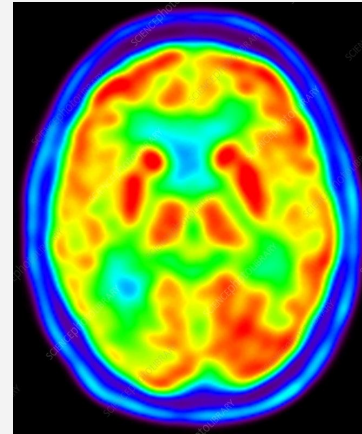
Pre-Processing

- Artefact (noise) removal
 - ECG, eye blinking, power-line noise
 - Blind Source Separation (BSS)
- Filtering (low-pass, high-pass, band-pass, notch)
 - Separate frequency bands
- Domain transform
 - Fourier transform (time to frequency domain)
 - Spectrogram
 - Periodogram



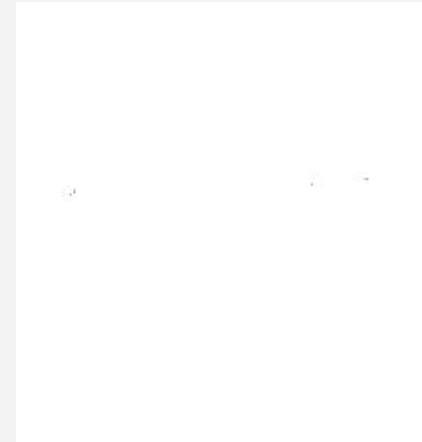
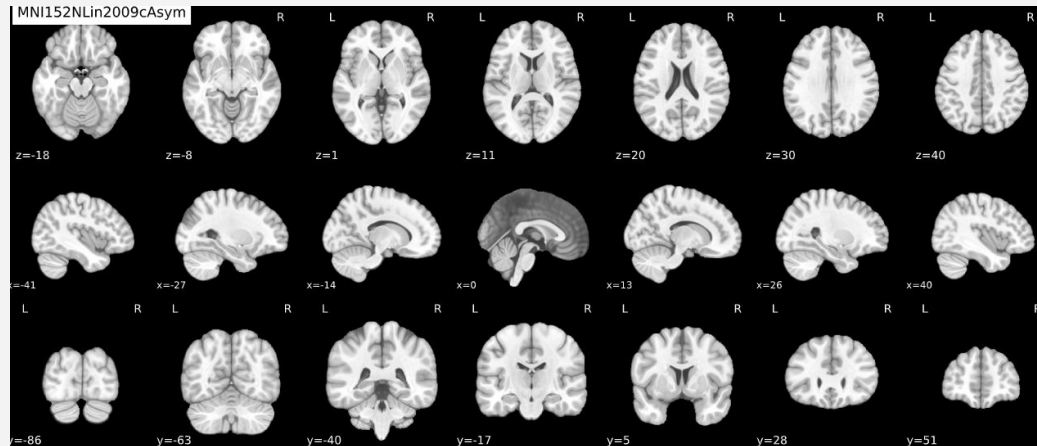
fMRI and PET

- fMRI: measures brain activity by detecting changes associated with blood flow
- Positron Emission Tomography (PET): visualize and measure changes in metabolic processes (blood flow, regional chemical composition, and absorption)



fMRI and PET (cont.)

- High spatial resolution and low temporal resolution
- Highly used in brain pathology (study of the causes and effects of disease or injury)
- Output is a 2D image of given plane of the 3D brain
- Scanners are not portable, expensive





Feature extraction methods for EEG

- Common spatial patterns (CSP)
- Autoregressive coefficients (AR)
- **Time-domain features**
- Zero crossing
- Waveform length (WL)
- Willison amplitude
- Inter-hemispheric amplitude ratio (IHAR)

Frequency-domain features

- Power spectral density (PSD)
- Peak frequency.
- Power spectrum deformation
- Frontal asymmetry
- Spectral coherence

Recreating brain

- Baby X

